

Code Guide

This document provides a technical overview of the E-Commerce Shopping Cart System to help developers understand the codebase, architecture, and logic.

1. Overview

The application is a pure Java command-line interface (CLI) e-commerce cart. It handles inventory management, cart operations, coupons, and checkout processes.

2. Project Structure

- `src/Main.java` : The entry point. Handles the interactive CLI menu and routes user inputs to cart operations.
- `src/Cart.java` : Manages the business logic of the shopping cart. Responsible for adding/removing items, updating quantities, applying coupons, calculating totals, and checkout.
- `src/CartItem.java` : Represents an individual line item in the cart, tracking the product and its quantity.
- `src/Product.java` : Data model for products, including ID, name, category, price, and stock quantity.
- `src/Coupon.java` : Data model for coupons, managing discount types (flat vs percentage), value, and expiry date.
- `src/FileHandler.java` : Manages I/O operations, specifically loading and saving `Product` and `Coupon` data to/from CSV files.
- `src/Utils.java` : Utility functions for the project.

3. Data Persistence

Data is loaded from CSV files upon startup and persisted back to the CSV file only upon a successful checkout.

- `data/products.csv` : Contains product definitions and stock levels. Handled via `FileHandler.loadProducts` and `FileHandler.saveProducts`.
- `data/coupons.csv` : Contains active coupons and their expiry details. Handled via `FileHandler.loadCoupons`.

4. Key Workflows

Adding to Cart

1. User enters Product ID and Quantity.
2. `Main.java` validates the product existence and quantity format.
3. `Cart.addItem` checks stock availability and creates/updates a `CartItem`.

Applying a Coupon

1. User selects a coupon.
2. `Cart.applyCoupon` verifies the coupon against the current date to ensure it is not expired.
3. Stores the applied coupon in the `Cart` instance for later calculation.

Checkout

1. `Cart.checkout` calculates final totals: Subtotal - Discount + Tax.
2. Deducts the purchased quantities from the actual `Product` stock.
3. Clears the cart.
4. `Main.java` detects a successful checkout, writes a receipt to `output/sample_outputs.txt`, and triggers `FileHandler.saveProducts` to update the CSV.